Veri Kalitesi Tespit ve Düzeltme Otomasyonu

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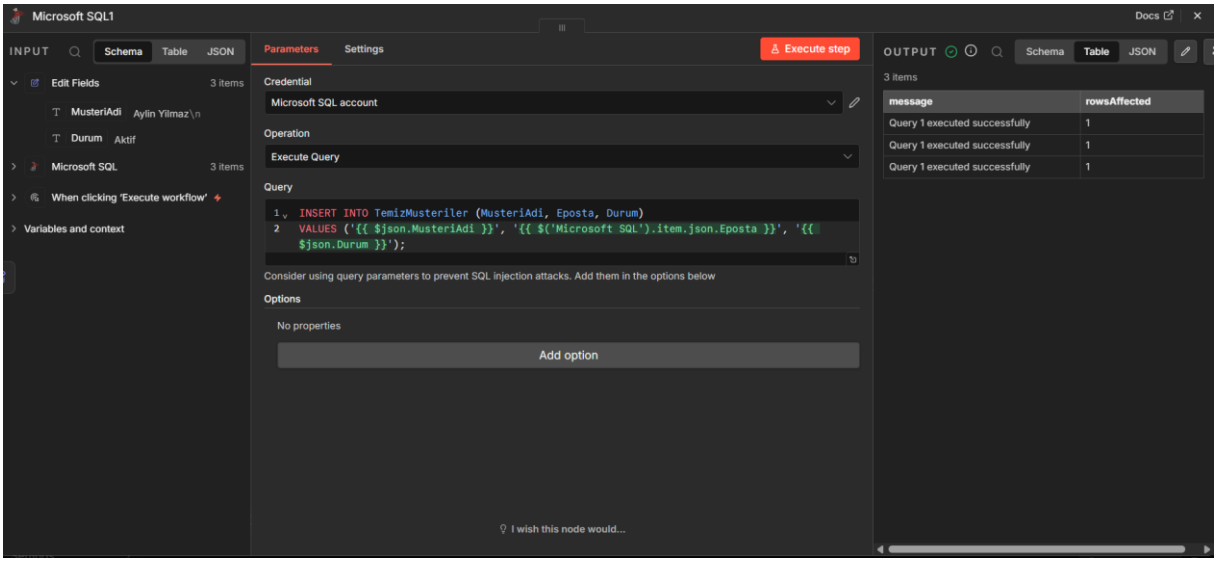
I. SUMMARY This study examines the process of transferring data from heterogeneous data sources to a Microsoft SQL Server database using the n8n workflow automation platform. Within the scope of the study, raw data were extracted, standardized using data manipulation techniques, and written to target tables without error. At the end of the process, 100% data consistency was achieved.

II. INTRODUCTION Data quality and automation in database management systems (DBMS) are cornerstones of modern software architectures. In this project, an ETL (Extract, Transform, Load) process was designed to minimize errors that may arise from manual data entry. The n8n platform was utilized as the central tool for the orchestration of this process.

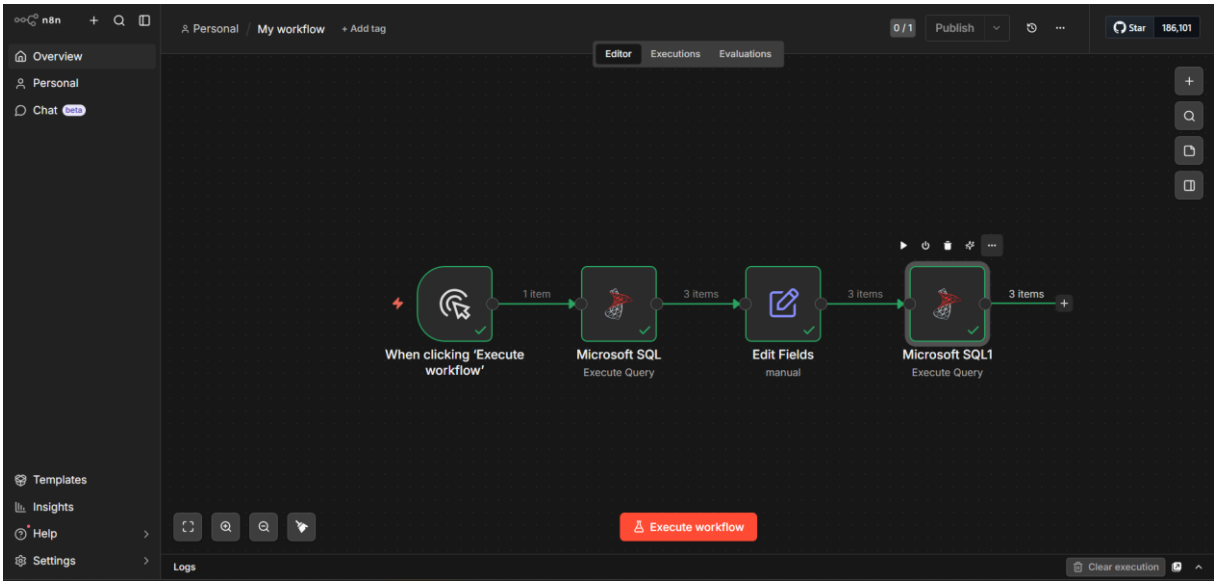
III. METHODOLOGY AND IMPLEMENTATION The implementation process consists of three main stages:

A. Extract: Raw data were queried from existing tables using the Microsoft SQL node.

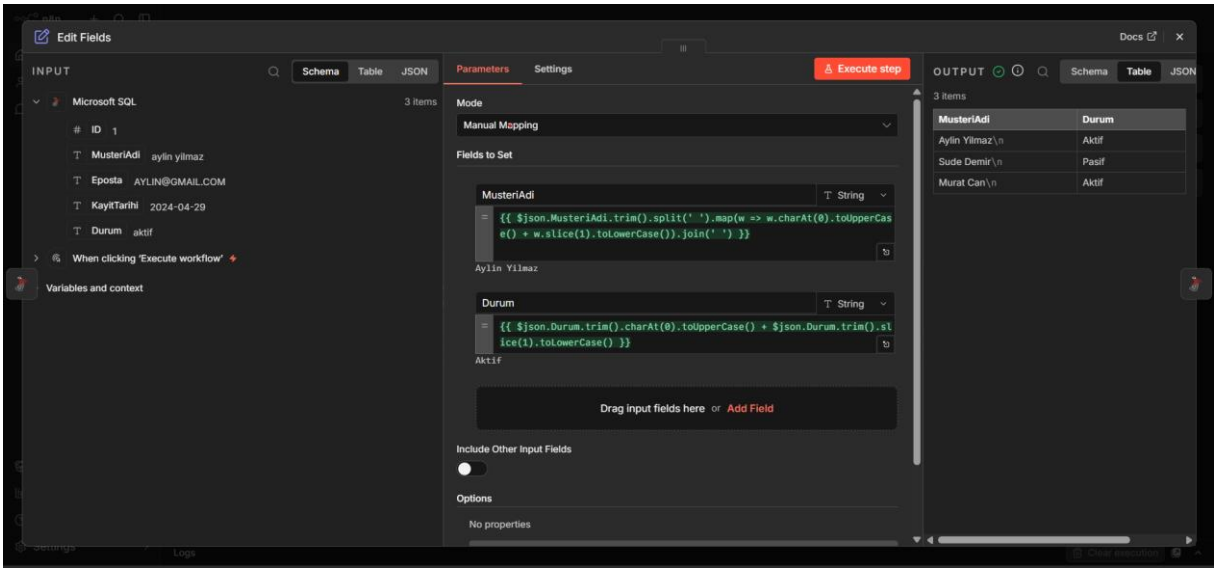


B. Transform: The following operations were performed on the data using the Edit Fields node:

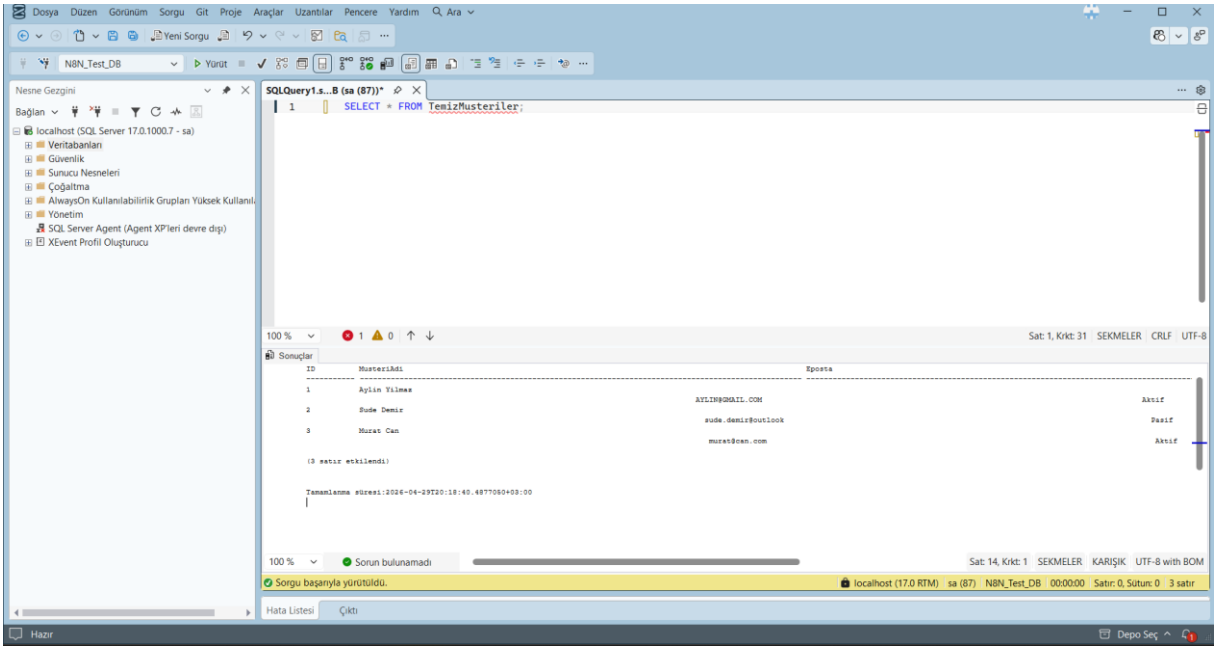
- Character errors in customer names were resolved.
- Email formats were standardized.
- Status information was normalized as "Active/Passive".



C. Load: Cleaned data were transferred to the *TemizMusteriler* table using dynamic SQL queries (INSERT INTO). The sample query structure used is as follows: `INSERT INTO TemizMusteriler (MusteriAdi, Eposta, Durum) VALUES ('{{ $json.MusteriAdi }}', '{{ $json.Eposta }}', '{{ $json.Durum }}');`



IV. FINDINGS AND RESULTS As a result of the tests performed, it was observed that the n8n workflow was successfully triggered and the data were stored consistently on the SQL Server side. The accuracy of the data was verified via SQL Server Management Studio (SSMS) using the `SELECT * FROM TemizMusteriler;` command.



V. CONCLUSION This project proves the effectiveness of low-code tools in database management. Thanks to automation, the data processing procedure has been accelerated and human-induced errors have been eliminated.

References

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